Energy

Track changes

1 Stores → transfers

Energy is conserved. Stores include kinetic, gravitational, elastic, thermal, chemical and nuclear.

Transfers: mechanically, electrically, by heating or radiation. Light and sound are not stores.

2 Calculate store changes

$E_k = \frac{1}{2}mv^2$; $E_p = mg\Delta h$; $E_e = \frac{1}{2}ke^2$. Use kg, m, m/s and N/m.

Spring: extension, not total length; stay within proportional limit. Doubling v or e quadruples energy.

Measure transfers

3 Power & efficiency



60 J useful + 40 J dissipated

$P = E/t = W/t$. Watts = joules per second. Same work in less time → greater power.

Efficiency = useful output / total input; ×100 for %. Dissipated energy is less useful, not destroyed.

4 Heating & insulation

$\Delta E = mc\Delta\theta$. Temperature change, not final temperature. Measure mass, energy and temperature rise.

Insulation: low conductivity and greater thickness reduce transfer. Compare cooling with controls.

Choose resources

5 Resources & reliability

Non-renewable: coal, oil, gas, nuclear. Renewable: biofuel, wind, Sun, tides, waves, hydro, geothermal.

Uses: heating, transport, electricity. Wind/Sun vary; tides are predictable but not continuous.

6 Evaluate a choice

Compare emissions, waste, habitat effects, cost, reliability and suitable sites.

A resource mix helps match demand. Science informs decisions; economic and social choices matter.

Stores & calculations

Follow the energy

1 Describe a change

Thrown ball: kinetic → gravitational; air resistance also warms surroundings.

Brakes: kinetic → thermal. Kettle: electrical transfer → water's thermal store.

In collisions, kinetic energy may decrease while total energy is conserved.

2 Conservation

Closed system: no net change in total energy. Include surroundings receiving transfers.

Dissipation spreads energy into less useful stores. "Wasted" does not mean destroyed.

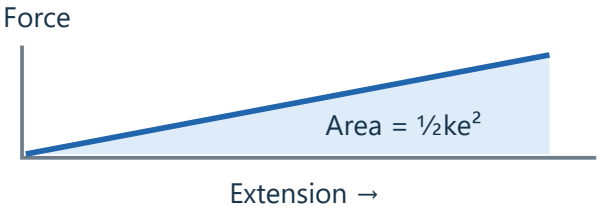
Use equations

3 Motion & height

$E_k = \frac{1}{2}mv^2$; $E_p = mg\Delta h$. Use given g. A 2 kg object at 3 m/s has 9 J.

Rearrange: $v = \sqrt{(2E_k/m)}$. Double speed → four times kinetic energy.

4 Springs



$E_e = \frac{1}{2}ke^2$ within proportional limit. e = stretched length − original length, in metres.

k = 200 N/m, e = 0.050 m → E_e = 0.25 J. Square the converted extension.

Reduce waste

5 Power is a rate

$P = E/t = W/t$; $E = Pt$. 1,800 J in 30 s → 60 W.

1 kW = 1,000 W; 1 kJ = 1,000 J. Powerful does not automatically mean efficient.

6 Efficiency

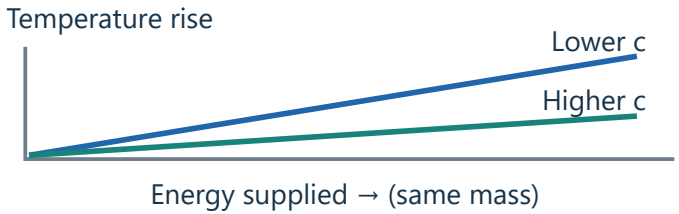
Useful output / total input, for energy or power. No unit; at most 1 or 100%.

350 J useful from 500 J → 70%; 150 J dissipated. Lubrication reduces friction and unwanted heating.

Heating & practicals

Link energy & temperature

1 Specific heat capacity



Energy to warm **1 kg by 1°C** without a state change. Units: J/kg°C.

ΔE = mcΔθ. Higher c → less warming for the same mass and input.

2 Worked example

0.50 kg water, c = 4,200 J/kg°C, warmed 20 → 40°C.

Δθ = 20°C; ΔE = 0.50 × 4,200 × 20 = **42,000 J.**

Convert grams to kg first.

Measure fairly

3 Measure c

Balance → mass; thermometer → rise; joulemeter or Pt → energy.

Insulate a block; improve heater/probe contact. Record initial temperature, heat, record rise.

c = E/(mΔθ). Repeat; keep electrical parts dry; avoid hot surfaces.

4 Explain error

Energy also heats equipment and surroundings. Using all input for the sample makes c too high.

Insulation reduces losses. Good thermal contact reduces lag. Energy vs temperature-rise gradient = mc ideally.

Limit cooling

5 Insulators

Thicker material or lower conductivity → slower conduction under matching conditions.

Trapped air reduces transfer; draught-proofing limits escaping warm air.

Payback time = cost / yearly saving, assuming constant savings.

6 Cooling investigation

Compare equal thicknesses; same containers, lids, water mass, starting temperature and room.

Measure temperature at equal intervals; repeat and average. Smaller drop in same time → better insulation.

For thickness: keep material fixed. Handle hot water safely.

Energy resources

Finite fuels

1 Fossil & nuclear fuels

Coal, oil, gas and nuclear fuels are finite. Fossil fuels support heating, electricity and transport.

Burning fossil fuels releases CO_2 and pollutants. Nuclear: reliable electricity, low operational CO_2, radioactive waste.

2 Costs beyond fuel

Extraction damages habitats. Nuclear plants need expensive construction/decommissioning.

Biofuel can regrow, but uses land and releases CO_2 when burned. Carbon balance depends on regrowth and production.

Renewable supply

3 Wind, Sun & waves

No fuel combustion during operation, but output varies with weather. Solar also varies with daylight.

Wind: visual/noise/wildlife concerns. Waves: harsh conditions and maintenance. Solar: area and siting matter.

4 Water & Earth

Hydro reservoirs can supply on demand but flood land; drought reduces supply.

Tides are predictable, not continuous; estuary impacts. Geothermal is steady at suitable sites, with drilling costs.

Make a judgement

5 Match supply to use

Heating, transport and electricity have different needs. Use a mix, storage and flexible backup.

Renewable means replenished as used; it does not mean always available or impact-free.

6 Use evidence

Compare power, reliability, cost, emissions, waste and habitats. Justify which matters most.

Read totals and percentage shares carefully. Scientific evidence cannot alone settle political or economic choices.